

VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi

Gokulam, 3<sup>rd</sup> Stage, Mysuru 570 002

Sixth Semester B. E. Examinations

NUMERICAL METHODS AND ITS APPLICATIONS

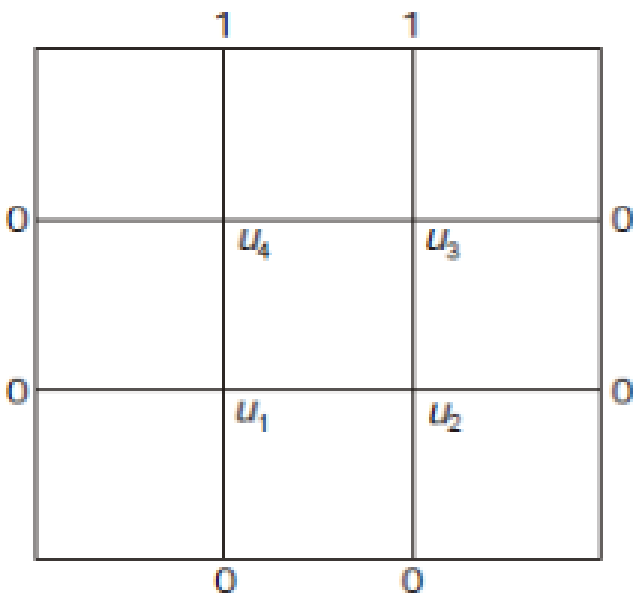
Duration: 3 hours  
100

Max. Marks:

INSTRUCTION TO STUDENTS

1) Answer one full question from each module.

| Q. No.     | Module-I                                                                                                                                                                                                                                                                                                   |   |    |    |    |     | Marks | BL | CO |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|----|----|-----|-------|----|----|
| 1. (a)     | Explain Inherent, Round-off and Truncation errors with an example. Also find the Absolute, Relative and percentage error in $x_A$ when $x_T=13$ and $x_A=0.333$ .                                                                                                                                          |   |    |    |    |     | 10    | 2  | 1  |
| 1. (b)     | Extract a quadratic factor using Bairstow method from the equation $x^4+4x^3+7x^2-22x+24=0$ with initial conditions $p_0=0.5$ and $q_0=0.5$ .                                                                                                                                                              |   |    |    |    |     | 10    | 3  | 2  |
| Module-II  |                                                                                                                                                                                                                                                                                                            |   |    |    |    |     |       |    |    |
| 2.(a)      | Find the inverse of the matrix $A=\begin{bmatrix} 3 & 4 & 2 \\ 1 & 1 & 3 \\ 5 & 1 & 2 \end{bmatrix}$ using matrix partition method and hence find the solution of the system of equations:<br>$\begin{aligned} 3x_1 + 3x_2 + 4x_3 &= 10 \\ 2x_1 + x_2 + x_3 &= 7 \\ x_1 + 3x_2 + 5x_3 &= 6. \end{aligned}$ |   |    |    |    |     | 10    | 4  | 2  |
| 2.(b)      | Solve the following system of equations by Cholesky method:<br>$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5. \\ 2x_1 + 8x_2 + 22x_3 &= 6. \\ 3x_1 + 22x_2 + 82x_3 &= -10. \end{aligned}$                                                                                                                        |   |    |    |    |     | 10    | 3  | 3  |
| Module-III |                                                                                                                                                                                                                                                                                                            |   |    |    |    |     |       |    |    |
| 3.(a)      | Fit a polynomial which takes the values                                                                                                                                                                                                                                                                    |   |    |    |    |     | 8     | 3  | 2  |
|            | x                                                                                                                                                                                                                                                                                                          | 1 | 2  | 3  | 4  | 5   |       |    |    |
|            | f(x)                                                                                                                                                                                                                                                                                                       | 0 | 10 | 38 | 96 | 196 |       |    |    |
| 3.(b)      | Find the approximate value of the integral $I=\int_0^1 dx \sqrt{1+x^2}$ using Romberg's method starting with the trapezoidal rule.                                                                                                                                                                         |   |    |    |    |     | 12    | 4  | 3  |
| OR         |                                                                                                                                                                                                                                                                                                            |   |    |    |    |     |       |    |    |

|                                                                                      |                                                                                                                                                                                                                   |   |   |   |    |    |   |   |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|----|----|---|---|
| 4.(a)                                                                                | Find $\frac{dy}{dx}$ at $x=1$ from the following table.                                                                                                                                                           |   |   |   |    | 8  | 3 | 2 |
|                                                                                      | x                                                                                                                                                                                                                 | 0 | 1 | 2 | 3  |    |   |   |
|                                                                                      | f(x)                                                                                                                                                                                                              | 1 | 3 | 7 | 13 |    |   |   |
| 4.(b)                                                                                | Evaluate the integral $I=\int_0^1 2x\sqrt{x^2+y^2} dx$ using Simpson's rule with $h= k=0.25$ .                                                                                                                    |   |   |   |    | 12 | 4 | 3 |
| Module-IV                                                                            |                                                                                                                                                                                                                   |   |   |   |    |    |   |   |
| 5.(a)                                                                                | Using the Householder's transformation reduce the matrix $A= \begin{bmatrix} 2 & 1 & 1 & 1 & 0 & 1 & 0 & 1 \end{bmatrix}$ into a tridiagonal matrix.                                                              |   |   |   |    | 8  | 3 | 2 |
| 5.(b)                                                                                | Using the Jacobi method, find all the eigenvalues of the matrix $A=\begin{bmatrix} 12 & -12 & 12 & -12 & 1 \end{bmatrix}$ . Iterate till the off-diagonal elements in magnitude are less than 0.0005.             |   |   |   |    | 12 | 3 | 2 |
| OR                                                                                   |                                                                                                                                                                                                                   |   |   |   |    |    |   |   |
| 6.(a)                                                                                | Find all the eigenvalues of the symmetric matrix $A= \begin{bmatrix} 12 & 42 & 12 & 42 & 1 \end{bmatrix}$ using Givens method.                                                                                    |   |   |   |    | 8  | 3 | 2 |
| 6.(b)                                                                                | Find all the eigenvalues of the matrix $A= \begin{bmatrix} 11 & 12 & 12 & 13 & 2 \end{bmatrix}$ using Rutishauser method. Iterate till the elements of the lower triangular part are less than 0.05 in magnitude. |   |   |   |    | 12 | 3 | 2 |
| Module- V                                                                            |                                                                                                                                                                                                                   |   |   |   |    |    |   |   |
| 7.(a)                                                                                | Solve the equation $u_{xx}+u_{yy}=0$ in the domain of the figure given below by Successive Over Relaxation method                                                                                                 |   |   |   |    | 10 | 3 | 2 |
|  |                                                                                                                                                                                                                   |   |   |   |    |    |   |   |
| 7.(b)                                                                                | Solve the equation $u_{xx}+u_{yy}=0$ in the domain of the figure given below by Alternating Direction Implicit method.                                                                                            |   |   |   |    | 10 | 3 | 2 |

# MODEL QUESTION PAPER: 2024-25

Course Code: BMANM561

[illegible]

**Course Outcomes:**

|            |                                                                                                                                                                                                                                                                                     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CO1</b> | <b>Understand</b> the basic concepts of Roots of a polynomial, Solution of a Linear system, Numerical differentiation and integration, Eigen values and Eigen vectors of Symmetric matrices, Boundary Value Problems in Ordinary and Partial differential equations. <b>(PO-1).</b> |
| <b>CO2</b> | <b>Apply</b> the concept of Linear system, Numerical differentiation and integration, Eigen values and Eigen vectors of Symmetric matrices, Boundary Value Problems in Ordinary and Partial differential equations. <b>(PO-1)</b>                                                   |
| <b>CO3</b> | <b>Analyse</b> the concept of Linear system, Numerical differentiation and integration, Eigen values and Eigen vectors of Symmetric matrices, Boundary Value Problems in Ordinary and Partial differential equations. <b>(PO-2).</b>                                                |
| <b>CO4</b> | Use <b>modern tool</b> to solve/visualize mathematical problems arising in Engineering <b>(PO-5).</b>                                                                                                                                                                               |

